SUPPLEMENTARY MATERIALS – S5

MROI Applicability Across Alternative Infrastructure Sectors

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S5.1 Purpose of Sectoral Extension

Sections S1–S4 demonstrated the MROI methodology using energy infrastructure archetypes.

This section evaluates applicability in three additional sectors:

- Transport infrastructure
- Building systems (HVAC and automation)
- IT infrastructure

The objective is not a full sectoral audit, but to demonstrate that the MROI equation and the JIM-30 correction are sectorally generalizable. The mathematical structure remains identical; only sector-specific parameters vary.

S5.2 Sector 1 – Transport Infrastructure

S5.2.1 Archetype Pairs

Archetype A – Closed-System Electric Bus Fleet (Parasitic Archetype) Manufacturer-exclusive battery replacement, server-dependent charging management, >90% imported components.

Archetype B – Modular Hybrid Public Transport Vehicle (Symbiotic Archetype) Standardized multi-supplier battery cells, open diagnostic protocol, mechanically serviceable components by local workshops.

S5.2.2 JIM-30 Scoring – Transport Sector

Dimension	Closed E-Bus	Modular Hybrid
I. Modularity	8/25	21/25
II. Parts Availability	4/25	20/25
III. Documentation	5/25	22/25

Dimension	Closed E-Bus	Modular Hybrid
IV. Local Repairability	6/25	21/25
Total JIM-30	**23/100 → VETO**	**84/100 → HIGH**

S5.2.3 MROI Calculation – Transport Sector

Parameter	Closed E-Bus	Modular Hybrid
ΔE_saved (kWh/year, diesel substitution)	85,000	92,000
grid_CO_2 (kg/kWh)	0.35	0.35
CO_2 savings (kg/year)	29,750	32,200
RACF units	56.56	61.22
JIM-30 (%)	23%	84%
Correction multiplier	×1.231	×1.048
EFU_input (corrected)	69.63	64.16
MROI	**–18.8%**	**+33.2%**
Classification	**PARASITIC**	**SYMBIOTIC**

Finding:
The transport sector shows an even larger MROI divergence than the energy sector (–18.8% vs. +33.2%). The closed battery architecture generates substantial infrastructure debt.

S5.3 Sector 2 – Building Systems

S5.3.1 Archetype Pairs

Archetype C – Smart Building Automation System (Parasitic Archetype)
Cloud-based control, proprietary sensor ecosystem, 5-year license cycles, monoblock controller units.

Archetype D – Modular Passive-House Mechanical System (Symbiotic Archetype)
Mechanical regulation with digital add-ons, open BACnet protocol, locally serviceable components, 30+ year mechanical lifespan.

S5.3.2 JIM-30 Scoring – Building Systems

Dimension	Smart Automation	Modular Passive
I. Modularity	9/25	23/25
II. Parts Availability	5/25	22/25
III. Documentation	4/25	24/25

Dimension	Smart Automation	Modular Passive
IV. Local Repairability	7/25	22/25
Total JIM-30	**25/100**	**91/100 → HIGH**

Note:
Dimension I for the smart system (9/25) falls below the automatic veto threshold (<10). The full calculation is presented for sensitivity demonstration.

S5.3.3 MROI Calculation – Building Systems

Parameter	Smart Automation	Modular Passive
ΔE_saved (kWh/year, 1,000 m² office)	18,500	22,000
grid_CO_2 (kg/kWh)	0.35	0.35
CO_2 savings (kg/year)	6,475	7,700
RACF units	12.31	14.64
JIM-30 (%)	25%	91%
Correction multiplier	×1.225	×1.027
EFU_input (corrected)	15.08	15.03
MROI	**–18.3%**	**+35.7%**
Classification	**PARASITIC**	**SYMBIOTIC**

Finding:
Although both systems deliver similar absolute energy savings, their MROI values diverge sharply.

MROI captures not only energy magnitude, but biophysical quality.

S5.4 Sector 3 – IT Infrastructure

S5.4.1 Archetype Pairs

Archetype E – Cloud-Based Municipal Data Service (Parasitic Archetype)
Hyperscaler data centers, continuous internet dependence, no hardware accessibility, escalating licensing costs, no local repair capability.

Archetype F – Local Edge Computing Municipal Server (Symbiotic Archetype)
On-site server hardware, open-source OS, locally maintainable, standardized hardware components, ensured data sovereignty.

S5.4.2 JIM-30 Scoring – IT Sector

Dimension	Cloud Service	Local Edge Server
I. Modularity	0/25	22/25
II. Parts Availability	0/25	21/25
III. Documentation	0/25	23/25
IV. Local Repairability	0/25	21/25
Total JIM-30	**0/100 → AUTOMATIC VETO**	**87/100 → HIGH**

Special Case – JIM-30 = 0
Cloud services externalize infrastructure resilience entirely. This zero score is a diagnostic outcome, not a methodological artifact.

S5.4.3 MROI Calculation – IT Sector

Parameter	Cloud Service	Local Edge Server
ΔE_saved (kWh/year, relative efficiency)	–3,200 (additional load)	+4,800
grid_CO_2 (kg/kWh)	0.35	0.35
CO_2 change (kg/year)	–1,120	+1,680
RACF units	–2.13	+3.19
JIM-30 (%)	0%	87%
Correction multiplier	×1.300	×1.039
EFU_input (corrected)	n/a	3.32
MROI	**Not interpretable – VETO**	**+24.1%**
Classification	**AUTOMATIC VETO**	**SYMBIOTIC**

Sectoral Specificity:
Cloud services exhibit negative net energy performance relative to local edge solutions, primarily due to cooling and redundancy overhead (IEA Data Centres Report 2023).

This is the only sector where MROI becomes structurally non-interpretable.

S5.5 Cross-Sector Comparison

S5.5.1 Summary Table

Sector	Parasitic Archetype	JIM-30	MROI	Symbiotic Archetype	JIM-30	MROI
Energy	Closed PV	25%	–3.0%	Modular Wind	92%	+27.1%
Transport	Closed E-Bus	23%	–18.8%	Modular Hybrid	84%	+33.2%

Sector	Parasitic Archetype	JIM-30	MROI	Symbiotic Archetype	JIM-30	MROI
Building	Smart Automation	25%	–18.3%	Modular Passive	91%	+35.7%
IT	Cloud Service	0%	VETO	Local Edge	87%	+24.1%

S5.5.2 Sectoral Patterns

Pattern 1 – Parasitic Archetypes: JIM-30 = 0–25%
Planned obsolescence, software lock-in, and manufacturer exclusivity consistently produce low scores.

Pattern 2 – Symbiotic Archetypes: JIM-30 = 84–92%
Open specifications and modular architecture consistently produce high scores.

Pattern 3 – Transport and Building Show Largest Divergence
Infrastructure debt accumulates more rapidly in these sectors.

Pattern 4 – IT as Structural Outlier
The digital sector uniquely combines low repairability with hidden energy overhead.

S5.6 Sector-Specific D Parameter Proposals

Sector	Suggested D Range	Rationale
Energy (baseline)	0.25–0.35	Moderate replacement cycle
Transport	0.35–0.50	Rapid technological obsolescence
Building systems	0.20–0.30	Slower aging, moderate lock-in risk
IT infrastructure	0.45–0.60	Fastest obsolescence cycle
Water systems	0.15–0.25	Mechanical dominance
Waste management	0.20–0.30	Mixed mechanical/digital

S5.7 Generalizability Conditions

MROI is generalizable to any infrastructure sector if three conditions are satisfied:

1. **ΔE_saved is measurable or modelable.**
2. **JIM-30 is applicable to physical system components.**
   (Full cloud abstraction triggers automatic veto.)
3. **RACF is regionally calibrated.**

If these conditions hold, the MROI equation applies without structural modification. Only the D parameter may require sectoral refinement.

S5 Status

Sectoral extension completed.
Four sectors demonstrate methodological generalizability.
Consistent structural patterns identified.

The Supplementary Materials series (S1–S5) is complete.

The full working paper (v1.3) is ready for Zenodo deposition and journal submission.